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Nuclear Engineering - Politecnico di Milano

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Course: Or other indication
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ABSTRACT

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Key-words: Key, Words, Here

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1 INTRODUCTION

1.1 Context

Italy is facing a significant energy transition, with the aim of achieving net-zero emissions by 2050. The residential sector plays a crucial role in this transition, as it accounts for $\sim 22\%$ of the total energy consumption and emissions [1].

The residential sector shows a high potential for decarbonization through the adoption of renewable energy sources such as solar photovoltaic (PV) systems and heat pumps (HP). However, this transition also introduces significant challenges in terms of monitoring and management, as it leads to a highly distributed production network that complicates national grid operations. Areas with dispersed populations and a high proportion of residential buildings often face particular difficulties during this transition. Solar photovoltaic systems are the primary choice for domestic renewable energy, but single-family houses are likely to feed a significant amount of their production into the grid, unlike apartment blocks of the same area where self-consumption is much higher.

In this context the role of nuclear energy is being reconsidered, especially with the advent of Small Modular Reactors (SMRs). Specifically, the present study focuses on the north-western part of the province of Varese, in the Lombardy region of Italy, where an hybrid system based on the integration of SMRs in a solar powered grid is being evaluated.

The integration of SMRs could provide a stable and reliable energy source to complement the intermittent nature of renewable energy sources such as solar PV systems, while also addressing the challenges of grid management and energy distribution in these areas.

1.2 Case Study

This study evaluates the technical and economic feasibility of integrating small modular reactors (SMRs) into the residential sector, considering the area's specific characteristics and the potential benefits and challenges of such implementation.

The proposed system consists of two SMRs, each with a nominal power output of 180 MWe, coupled with a secondary system capable of producing hydrogen through high-temperature steam electrolysis (HTSE). This system also includes synthetic gas production via CO2 methanation, which feeds into the gas mains and powers a small gas turbine with a nominal electrical output of 47 MWe. The electricity generated by this integrated system along with the residential PV production is designed to meet the entire demand of the service area. Additionally, the plant is configured to handle demand peaks and enhance profitability and adaptability to various energy market requirements.

1.2.1 Reactor

Our plant consists of two SMRs installed in Ispra, where the Joint Research Center (JRC) is located. The JRC is a research facility of the European Commission which has been chosen since it represents an already nuclearized site. This suggests the possibility of an easier pathway towards the construction of a future nuclear plant.

Moreover, the facility is home to the ESSOR reactor, currently facing decommissioning. The installation of at least one of the two SMRs could take place, technically, within the ESSOR reactor building. The containment was oversized due to the particular use case of the ESSOR reactor:

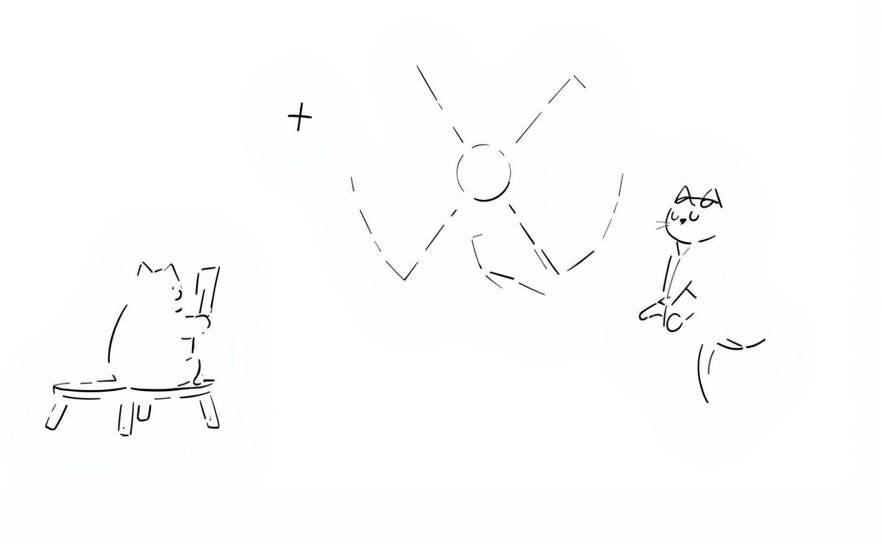


Figure 1: Simple schematic of the proposed hybrid plant integrating SMRs, hydrogen production, methanation, and residential PV.

primary and secondary circuits, as well as several laboratories are accommodated within the main building.

1.2.2 Secondary Plant

The secondary plant is designed for dual-purpose hydrogen utilization. Its primary function is to produce hydrogen through high-temperature steam electrolysis (HTSE), with a constant output allocated for market supply. Additionally, the plant uses the remaining hydrogen to produce synthetic gas via CO2 methanation. This synthetic gas production serves two main purposes: it fuels the gas turbine and creates a surplus that can be stored to reduce the region's dependence on fossil fuels, thereby supporting decarbonization efforts.

1.3 Objectives

The study aims to evaluate both the technical and economic feasibility of the proposed installation.

From a technical perspective, the system's capacity to meet energy demands and adapt to transient conditions will be assessed. From an economic standpoint, the total plant costs will be analyzed to determine the conditions required for achieving a positive net present value (NPV) within a reasonable timeframe.

The evaluation will consider the entire plant operating over a time span of 60 years. The evolution of residential energy and building performance is simplified into four scenarios, each representative of a 15-year period:

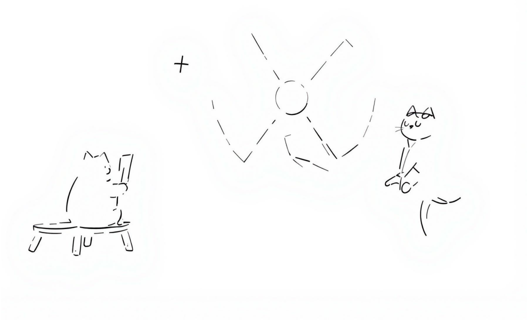
- Years 1-15: Current state of the residential sector in the area

- Years 16-30: There is still a significant share of fossil fuel heating systems in the residential sector

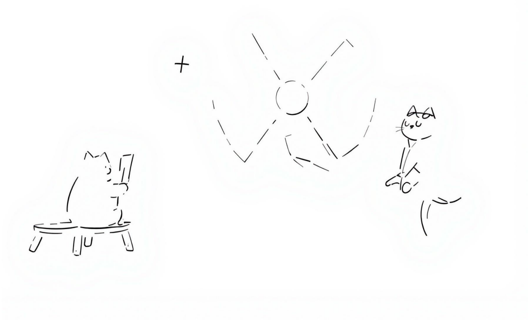
- Years 31-45: A middle ground between the optimistic and pessimistic scenarios, with a balanced mix of electrification and fossil fuel systems

- Years 46-60: Residential sector is mostly electrified with highly efficient homes

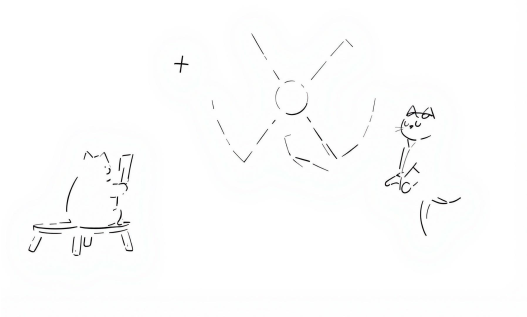
Given the constraints of the available data, the assessment will consider the plant to be operational as of the current year (2025). At the conclusion of the study, considerations will be made regarding a more realistic starting time. The plan is to abstract from a specific starting time by considering capital and operational costs relative to each other.



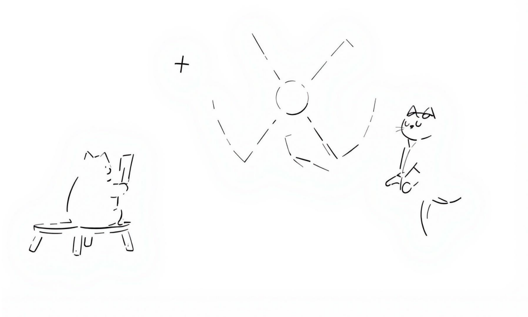
(a) 1-15 Years Scenario



(b) 16-30 Years Scenario



(c) 31-45 Years Scenario



(d) 46-60 Years Scenario

Figure 2: Distribution of energy classes in the residential sector for the four scenarios.

The effectiveness of the hybrid system is evaluated with respect to the following key aspects:

- The investment cost to make the plant profitable

- The decarbonization impact

- The effectiveness of integration of the several components of the plant

1.3.1 Decarbonization Impact

The decarbonization impact will be assessed through an analysis of CO_2 emission reductions achieved by implementing the hybrid system. A comparative approach will be adopted, examining emissions from both the current residential sector energy mix and those projected for the proposed system across various scenarios.

The reduction will be quantified based on two primary factors:

- The volume of conventional gas replaced by CO_2-neutral synthetic gas
- Changes in the overall energy mix composition

Both aspects will be evaluated using grams of CO_2-equivalent per kilowatt-hour ($\frac{gCO_2-eq}{kWh}$) as the measurement unit. The decarbonization impact will ultimately be expressed as a percentage variation from the current state of the energy mix.

2 DATA COLLECTION

At the basis of this work there is the need to have a realistic energy demand profile for the residential sector in the region under study. Given our focus on the residential sector, the overall energy demand can be seen as the sum of the heating/cooling contribution + all the other electricity consuptions for appliances and lightning. Several semplifications will be made through the following derivation but we point out any possible improvment that could be made but was considered out of the scope of this work.

2.1 Approach

2.1.1 Introduction

In Italy, an APE (Attestato di Prestazione Energetica) is a document that certifies the energy performance of a propriety unit (*unità immobiliare*). This certification must be issued by a qualified technician and is mandatory for all buildings that are sold, rented or refurbished.

The APE classifies buildings in energy class, a letter grade (from A4 to G), that indicates the energy performance of the building, with A4 being the most efficient and G being the least efficient. The certificate includes many informations about the building among which it is possible to find information about geometry, plants, materials as well as the use of the building (commercial, residential, industrial, public, etc.).

Each unit refers to a distinct and identifiable part of a real estate property, such as an apartment, a house, or a commercial space, that can be individually owned, sold, or rented. For instance, a building with three apartments will have three separate APEs, one for each apartment. For this reason we can safely assume that each unit is the residence of a single family.

2.1.2 Methodology

The plan is to determine the demand profile of electricity in the region $P_{e,\text{tot}}$ from a distribution of energy classes. To do so it is necessary to first determine the reference demand profile for each energy class $P_{e,\text{ref},i}$, where i is the energy class.

For each energy class there is need to determine:

- Heating thermal demand $\dot{Q}_h$
- Cooling thermal demand $\dot{Q}_c$
- Domestic hot water thermal demand $\dot{Q}_{dhw}$

For each energy class we also need the share of heat pumps $\chi_{HP,i}$ to differentiate between electrified and fossil based heat generation. The primary energy demand (electricity or fossil fuels) are obtained in a simplified manner with average efficiencies.

Regarding the electricity production from photovoltaic system we need to determine the share of buildings with photovoltaic systems for each class $\chi_{PV,i}$ and the average size of the photovoltaic system $P^{+}_{e,i}$ for each energy class, this last was differentiated between households with and without an heat pump. A reference profile of the PV production for the region is obtained and then weighted on the simulated scenario.

Another electricity contribution is the one from appliances and lighting, $P_{e,other}$ which is assumed to be equal for all classes.

$$
\begin{aligned}
P_{e,\text{ref}} = \sum_{i=\text{A4}}^{\text{G}} \Bigg[& \chi_{\text{HP},i} \cdot \left(\frac{\dot{Q}_{h,i}}{\text{COP}} + \chi_{\text{cooling}} \cdot \frac{\dot{Q}_{c,i}}{\text{EER}} \right) \\
& - \chi_{\text{PV},i} \cdot \left(\chi_{\text{HP},i} \cdot P_{\text{PV,w/HP}} + (1 - \chi_{\text{HP},i}) \cdot P_{\text{PV,w/oHP}} \right) \\
& + P_{e,\text{other}} \Bigg]
\end{aligned}
\tag{1}
$$

Then, this can be multiplied by the number of buildings in the region N_{UI} to obtain the total demand $P_{e,\text{tot}}$.

$$
P_{e,\text{tot}} = N_{\text{UI}} \cdot P_{e,\text{ref}} \tag{2}
$$

2.1.3 Data Sources

From the CENED database [2] we extracted the information about the relevant municipalities of the region, which we derived based on the map of primary substations from the national grid operator (GSE) [3]. The data was also filtered to include only residential buildings, excluding commercial and industrial ones. Furthermore, we filtered unhealthy data points at every step of the process, such as buildings with areas larger than $1000m^2$ (mostly releated to errors in unit of measurement or lack of care in the production of the APE). This was necessary given the large amount of data from the complete regional database. This way an easier to manage csv file was obtained. The number of buildings was computed from the ISTAT database [4] by extracting the population of the municipities and dividing by the average household size in the province.

2.2 Heating and Cooling Demand

For each energy class we have taken a real example of a residential building from the region under study and computed the hourly thermal demand for heating and cooling thoughout the year. This is obtained by means of a professional software (TERMOLOG) that allows for dynamic simulation of the energy system of the building. This computation is based on the UNI EN ISO 52016 standard. We used one reference building per energy class all with the same utlization profile. The output of the calculation is an hourly profile of the thermal demand for heating and cooling.

2.3 Domestic Hot Water Demand

The domestic hot water demand is assumed to be equal for all energy classes. It has been computed based on the consuption profile depicted in [5], as a reference we choose the profile of swizerland since it is similar to the one of the region under study. It is then scaled based on the average number of people per household in the region, which is 2.25 [4]. This thermal demand is then summed to the heating demand profile.

2.3.1 Normalization of the Data

Given that our data is based on real building we had to normalize the data to obtain a reference value for each energy class. This is done by taking into account the userful heated area of the building and the EP,nren (Primary Energy Demand for non-renewable energy) value of the APE. The EP,nren is the value used to determine the energy class of the building and is defined as energy per unit of area per year. Therefore we multiply the EP,nren by the useful heated area of the building to obtain the total primary energy demand for non-renewable energy. Convertion factor for each energy class are taken as the ratio between said value obtained by the averages extracted from the CENED database [2] and that for our reference buildings.

$$C = \frac{\text{EP,nren}_{\text{CENED}} \cdot A_{\text{r,CENED}}}{\text{EP,nren}_{\text{REF}} \cdot A_{\text{r,REF}}} \tag{3}$$

2.4 Heat Pump distribution and Cooling Demand

We identified the share of heat pumps per energy class. We also determined the share of buildings that have a cooling system installed, which as of today comes up to 9.58%. This is a very low share and is to be expected as the region under investigation is quite chill during summer.

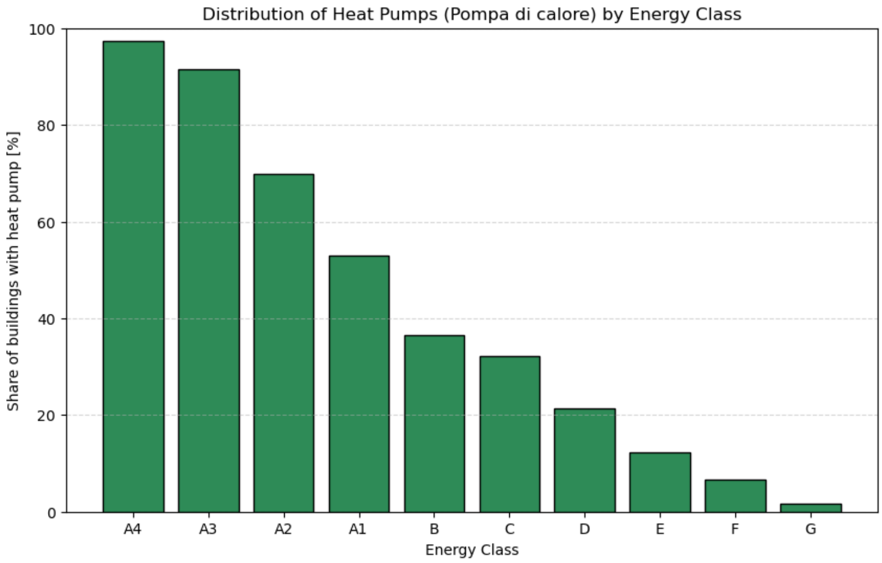


Figure 3: Distribution of heat pumps across energy classes.

2.5 Photovoltaic

2.5.1 Photovoltaic Distribution

We identified the share of building that have photovoltaic systems installed per energy class.

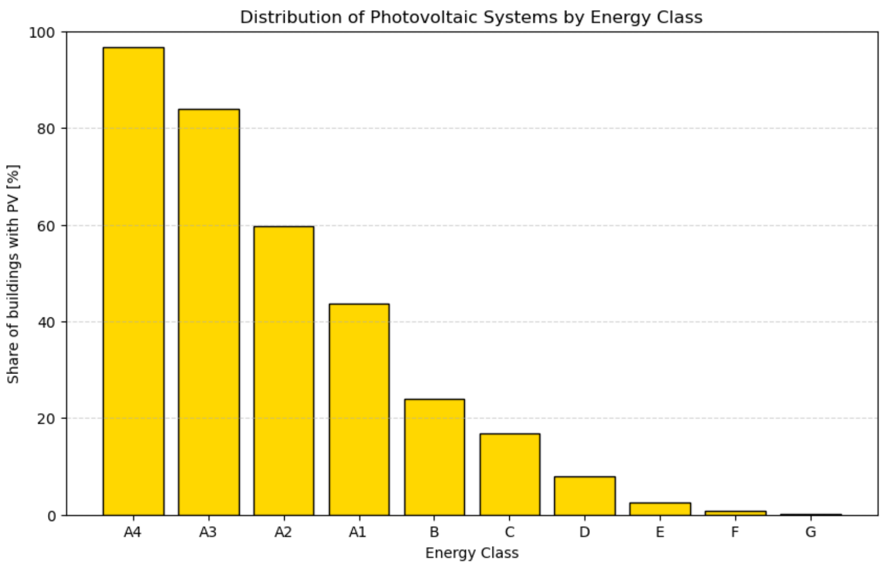
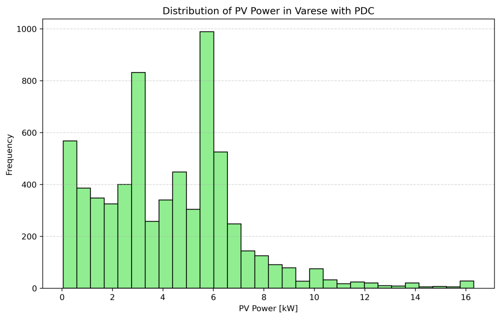


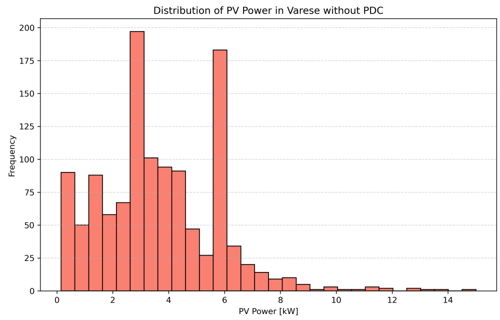
Figure 4: Distribution of photovoltaic systems across energy classes.

2.5.2 Photovoltaic System Size

Moreover we then divided the data into two groups, those with an heat pump and those without. From these we plotted the distribution of the size of the photovoltaic system installed. And computed the average (of the P_99)



(a) Photovoltaic system size distribution for buildings with heat pumps.



(b) Photovoltaic system size distribution for buildings without heat pumps.

Figure 5: Distribution of photovoltaic system sizes based on heat pump presence.

The averages values are: 4.30kW and 3.70kW for the sistem with and without heat pumps respectevely.

2.5.3 Reference Production

We have to obtain a reference value of the photovoltaic production profile. First, the list of municipalities is fed to the Nominatim API [6] to obtain the coordinates of the center of each municipality. Then, we use the PVGIS API [7] to obtain the hourly production profile for each municipality. We fixed the angle to an average $22°$ and a plant size of $1kW$. For each municipaly we obtain a profile every $10°$ of the azithmut angle, from $0°$ to $360°$. Those are then weighted to a gaussian distribution to represent that it is prefered to point the photovoltaic system to the south. This gives us a reference photovoltaic production profile for each municipality $P^+_{e,m}$, where m is the municipality. Then we can obtain the reference production profile for each energy class $P^+_{e,ref}$ by weighting the production profile of each municipality on the population in that municipality [4].

2.6 Other Electricity Demand

Electrical demand for appliances and lighting is modeled equal for all classes. The definition was manual, hour by hour for an average day in the following "seasons":

- Winter Weekday
- Winter Weekend
- Summer Weekday
- Summer Weekend

- Winter Holiday
- Summer Holiday

Moreover we added an absence factor to take into account when people go on holiday massively during august.

The profile is defined by considering the following contributions:

- standby and always on appliances
- lighting
- appliances

During the summer the need for a cooling fan was added. Moreover we added a parameter to rapresent the share of buildings with an induction system installed.

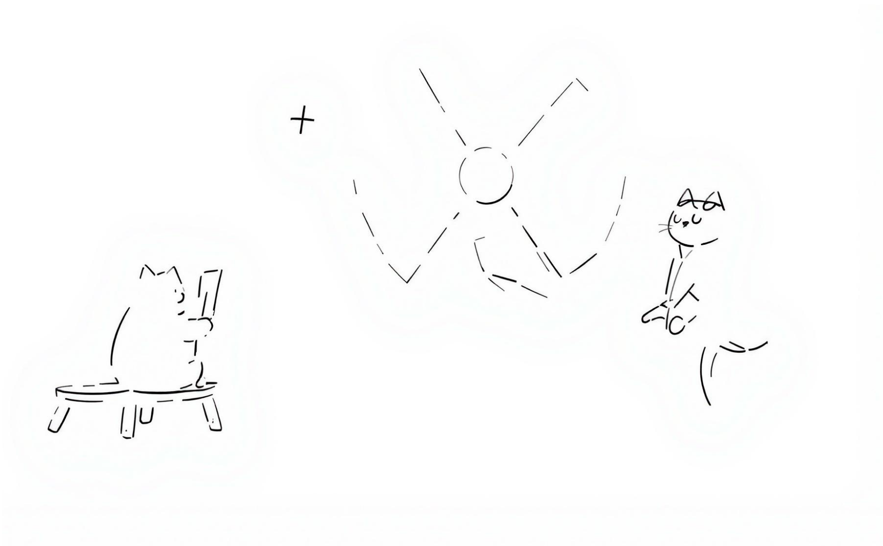


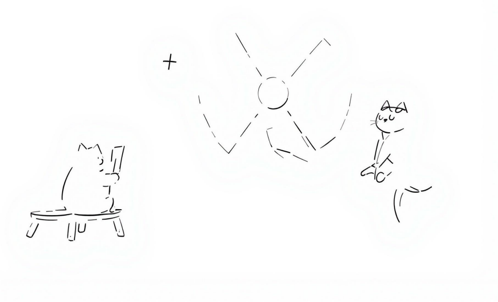
Figure 6: Reference hourly profile through the year.

2.7 Validation

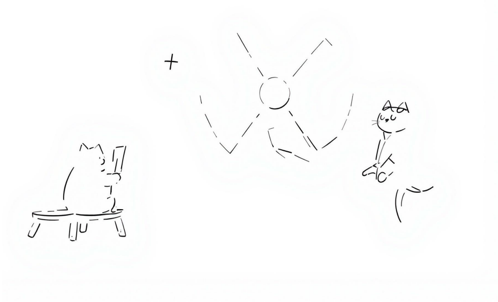
To verify our method is acceptable we compare the total demand by the data available online, in particual we compared it to the average annual deamnd, in the residential sector for the province of Varese [8], which happends to be $1931kWh$ in 2023, with a decreasing trend in the last years ($2016kWh$ in 2022 and $2119kWh$ in 2021).

2.8 Demand Distribution

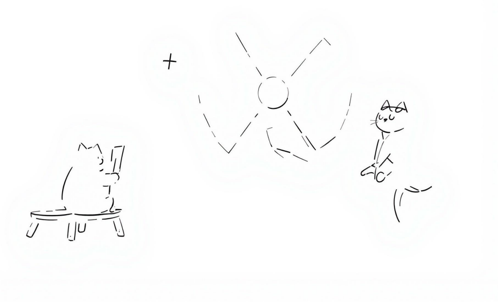
To better visualize the behaviout of the demand we plot the distribution of the demand hour-by-hour for each scenario.



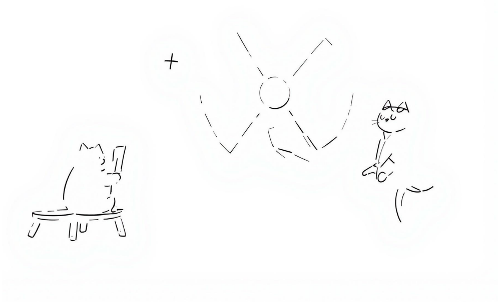
(a) Scenario 1



(b) Scenario 2



(c) Scenario 3



(d) Scenario 4

Figure 7: Distribution of the demand hour-by-hour for four scenarios.

3 TECHNICAL PLANT MODELLING

The whole plant was modeled in Dymola, a technical software using the Modelica language for the dynamic analysis of complex systems. To meet the overall energy demand of the target region, the plant design combines two 177 MW_e Small Modular Reactors (SMRs) with fifteen High Temperature Steam Electrolyzers (HTSEs) and a Methanation Reactor producing almost CO_2-neutral syngas for a small Gas Turbine (47 MW_e). The TANDEM project library [9] was widely used to model the enritre plant, with the exception of the methanation reactor, which was modeled from scratch in Dymola.

3.1 Main assumptions and plant layout

The number of reactors and HTSE modules is based on the following considerations:

- The minimum number of installed HTSE modules required has to sustain the Gas Turbine operation at the nominal power, guaranteeing a sufficient methane production.
- The maximum number of installed HTSE modules depends on the overall energy excess of the system in the least demanding hour of the year.
- The overall energy excess of the system depends on the number of SMRs.

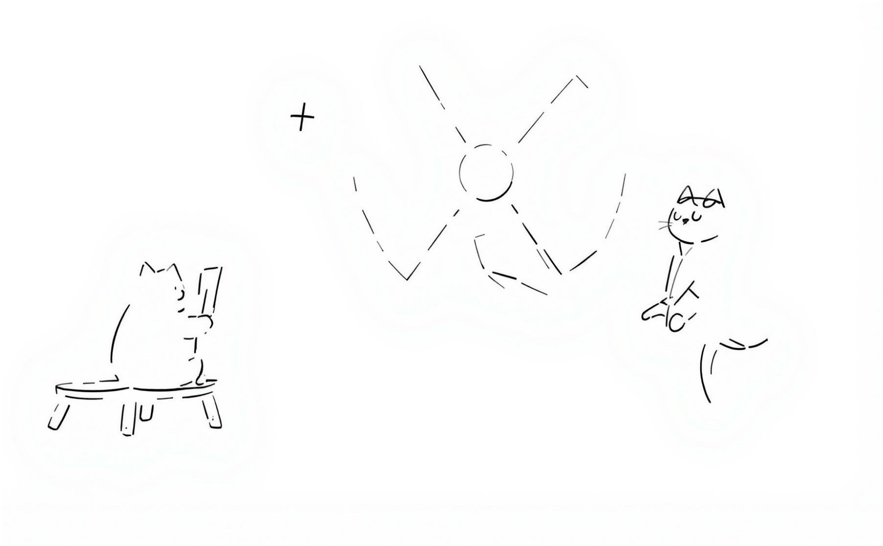


Figure 8: Simple schematic of the proposed hybrid plant in Dymola.

In order to find the optimal number of SMRs and HTSE modules which would minimize the overall cost of the plant, while meeting the energy demand, an optimization problem would be required. For the purpose of this study the optimization is not performed.

The choice was then made on the number of HTSE modules, simply considering the average between the minimum number of modules required (12) and the maximum allowed (18, with two SMRs). This results in a total of 15 HTSE modules. Accordingly, as the HTSE modules require a significant amount of electricity to operate, the number od SMRs was set to two. This allows for an efficient operation of the whole plant, with the need to import electricity just in less than 10% of the hours in a year.

However, to reduce the computational burden of the simulation in Dymola, the system is modeled using a single SMR and a single HTSE module. The actual electricity output from the SMR and hydrogen production from the HTSE are then scaled by multiplying the simulated values by the respective number of units within the model. These scaled values are used to evaluate the overall energy balance and to feed the syngas production module.

The SMRs supply electricity to the grid, as well as both electricity and the required heat to the hydrogen production unit. The HTSEs generate hydrogen from water using the thermal and electrical energy provided by the SMRs.

At each time step, 10% of the produced hydrogen is assumed to be stored temporarily in a buffer for future sale. The remaining 90% is sent to the methanation reactor, where it reacts with carbon dioxide to produce methane.

The profile and magnitude of the future hydrogen demand is not known: the only purpose of this simplified assumption is to allow the technical evaluation of the plant's operation, while considering a reasonable amount of hydrogen to be stored and sold. An excessive hydrogen storage would not be feasible in a real-world scenario, as it would require significant investments in storage infrastructure.

The syngas produced by the methanation reactor is ultimately used as fuel in the gas turbine, while any surplus is stored in a buffer tank. The stored syngas can later be used to cover the residual fossil fuel demand of the region (i.e., the remaining heat demand not met by other sources), substantially contributing to the decarbonization of the energy mix.

3.1.1 SMR

The TANDEM library [9] was used to model the SMR, employing a simplified representation of a Small Modular Reactor. Despite being a simplified model, the overall system is quite detailed and includes various components, allowing for a realistic simulation of the reactor's behavior. It captures the dynamics of the primary and secondary loops, the turbines, the condenser, and the associated control systems.

The selected model is part of the POLIMI section of the SMR branch in the TANDEM library. Specifically, the "Test_BOPfluid_NSSS" [9] is a test model of the 177 MW$_e$ light-water cooled Small Modular Reactor (SMR), based on the European SMR (E-SMR) design as proposed by the Euratom ELSMOR project [10]. The model is based on elementary components taken from the ThermoPower library [9].

The employed test model was originally designed for cogeneration, providing both electricity and thermal energy. To support this dual purpose, the Balance Of Plant features three steam extraction points—High Pressure (HP), Intermediate Pressure (IP), and Low Pressure (LP)— which allow heat to be drawn from different stages of the turbine. In the present configuration, the heat required for the HTSE is extracted via the IP tap, immediately downstream of the HP turbine stages. The HP and LP extraction lines are disabled and manually set to zero, as they are not utilized in this setup.

For the purpose of this study, the detailed modeling of the primary loop is not discussed, as the focus is on the secondary loop and its interaction with the HTSE modules. However, the detailed explanation of the reactor's core and primary loop modelling can be found in the TANDEM library documentation [10].

Some useful specifics of the SMR model are:

SMR specifics

- Core Thermal Power: 540 MW_{th}
- Core Inlet Temperature: 300 °C
- Core Outlet Temperature: 324.5 °C
- Primary Loop Pressure: 150 bar
- Nominal Coolant Mass Flow Rate: 3700 kg/s
- Thermodynamic Efficiency: 0.33
- Electrical Power Output: 177 MW_e
- Secondary loop: Rankine cycle with multiple turbine stages
- Steam extraction points pressure and temperature [10]:
  - HP: 45 bar, 240 °C
  - IP: 7.5 bar, 150 °C
  - LP: 0.8 bar, 80 °C

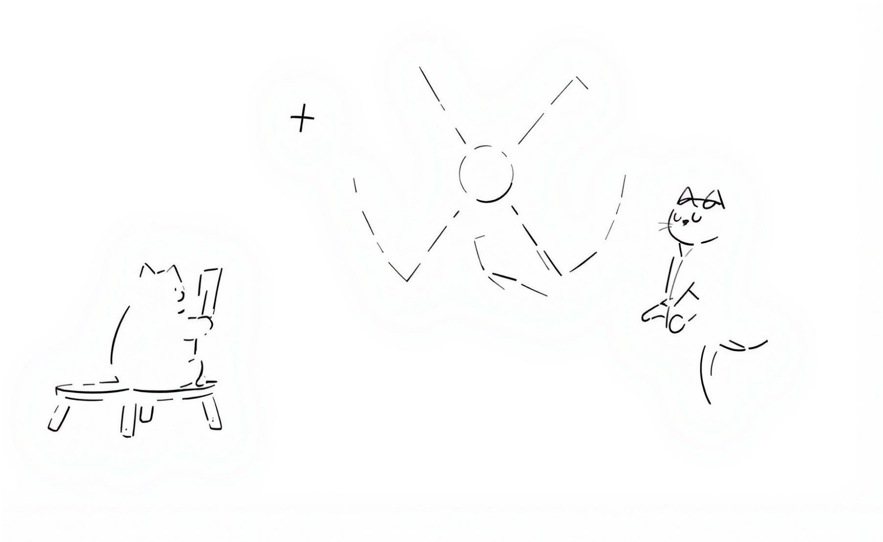


Figure 9: Simple schematic of the selected SMR model in Dymola.

3.1.2 H2 Plant

The hydrogen production plant is also modeled using the TANDEM library [9], which provides a detailed representation of a High Temperature Steam Electrolyzer (HTSE) based on the ThermoSysPro library. This Solid Oxide Electrolyser Cell operates at high temperature, and is designed to produce hydrogen from water by utilizing both thermal and electrical energy supplied by the SMR. The model captures the dynamic behavior of the HTSE and simulates the hydrogen production process accordingly.

Water goes through several heat-exchangers to reach the target temperature of 750 °C as steam:

- A pre-heater.
- A boiler, fed by the tap-steam extracted from the SMR BOP (at 150 °C).
- A second pre-heater, fed by the recovery heat from unconverted steam.
- An electrical heater.

Each electrolyser's stack then uses electricity to act on the high temperature steam and produce hydrogen.

The model receives as input a target hydrogen production rate - further defined in the demand section - and the thermal energy extracted from the SMR's steam tapping point.

The electricity consumption is instead related to the hydrogen production rate, according to the relationship $\dot{m_{H2}} = \frac{P_{e,available}}{214} + 0.197775$, which was found by running a sweep simulation of the HTSE model in Dymola.

A key modeling assumption is that the heat input to the HTSE corresponds to the `Real` value of the heat exchanged in the ideal Heat Exchanger of the Intermediate Pressure (IP) tap loop, once the tap-steam flowrate is defined.

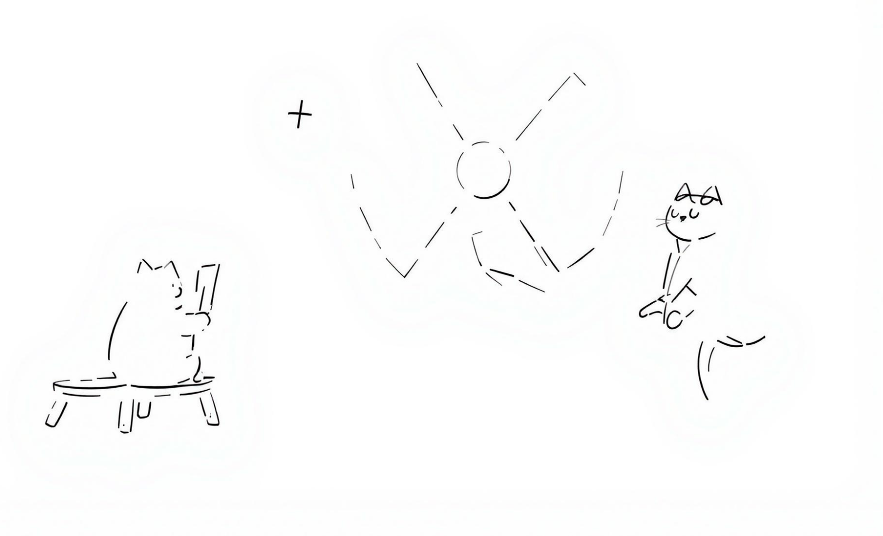


Figure 10: Simple schematic of the Intermediate Pressure (IP) tap loop.

The tap-steam flowrate is defined in the Control Room section, where the setpoint for the steam extraction is computed based on the target H_2 production. Again, the relationship between the tap-steam flowrate and the hydrogen production rate is defined by running a sweep simulation of the HTSE model in Dymola.

Based on the thermal input, along with several operational parameters, the HTSE model calculates the corresponding hydrogen production, resulting in a computed mass flow rate of H_2. The estimate of the energy consumption associated with hydrogen production is also computed within the model, and subsequently included in the overall energy balance of the system.

Some key characteristics of the HTSE include:

- Operating Temperature: 750 °C
- Electricity Consumption: related to the hydrogen production rate
- Nominal Production Rate: 0.1 kg/s of H_2 per module
- Number of Modules: 15

The H_2 plant is equipped with a buffer tank used to temporarily store the portion of hydrogen intended for sale, while the remainder is sent directly to the methanation reactor.

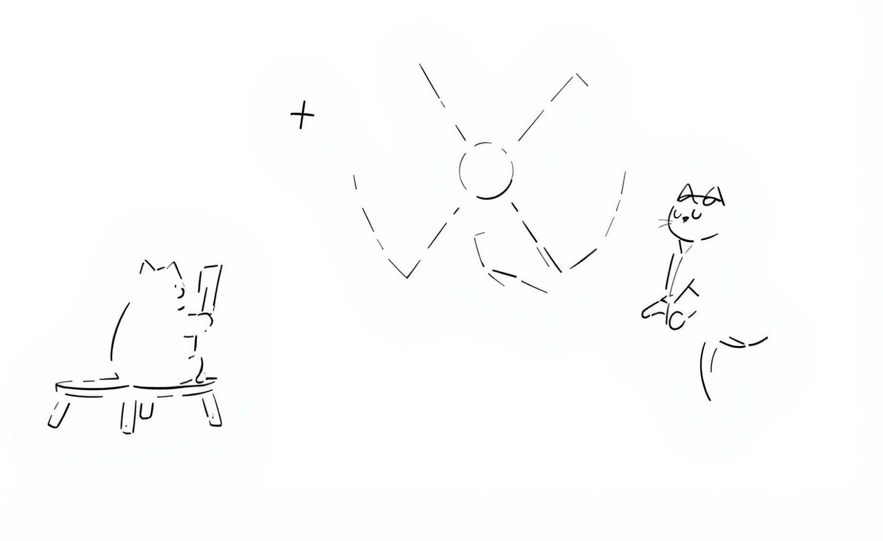


Figure 11: Simple schematic of the H_2 plant model in Dymola.

As the model is designed to simulate the operation of a single HTSE module, the target hydrogen production rate (and the effective output production rate) is obtained by dividing (and multiplying) the total input (and output) value by the number of modules (15), through a simple "Gain" component.

3.1.3 Methanation Reactor Modeling Approach

The methanation reactor was modeled to simulate the conversion of hydrogen (H_2) and carbon dioxide (CO_2) into methane (CH_4) and water (H_2O) through the Sabatier reaction:

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O \qquad \Delta H = -165.1 \text{ kJ/mol} \tag{4}$$

To estimate the amount of CH_4 produced from a given reactant feed, a dimensionless parameter, $conversionEfficiency$, was introduced. This parameter (default value: 0.78) represents the maximum chemical efficiency of catalytic methanation, as reported by Götz et al. [11]. It defines the fraction of the limiting reactant (either CO_2 or H_2) that is effectively converted into CH_4, accounting for real-world inefficiencies and the incomplete conversion typical under practical temperature and pressure conditions.

The molar output flows are then determined based on reaction stoichiometry:

$$\begin{aligned}
\dot{n}_{\text{CH}_4} &= \dot{n}_{\text{lim}} \cdot \eta \\
\dot{n}_{\text{H}_2 O} &= 2 \cdot \dot{n}_{\text{CH}_4} \\
\dot{n}_{\text{H}_2} &= \dot{n}_{\text{H}_2,\text{in}} - 4 \cdot \dot{n}_{\text{CH}_4} \\
\dot{n}_{\text{CO}_2} &= \dot{n}_{\text{CO}_2,\text{in}} - \dot{n}_{\text{CH}_4}
\end{aligned}$$

The model is reported in the appendix, along with the other components developed from scratch in Dymola.

Stechiometry and Output Mixture Given that the reaction is not ideal if we respect the Stechiometry (1:4) the output mixture would result in a lower LHV then the one of natural gas. To improve the energetic and economic value of our product, we decided to run an over-stechiometric reaction, specifically 1:8 parts of CO_2 and H_2. This means that the output mixture will be:

- 41.1 CH4
- 30.7 H2
- 28.2 CO2

We obviously assume all water is discarded from the product. The resulting LHV is 57.4359.

CO_2 **Usage** Based on our mixture we obtain a mass flow rate of syntetich gas of 2.6325, given the stechiometry the inpyt of CO2 is 3.7 therefor we can deduct that to produce a kg of SG we need 1.41 kg of CO2.

3.1.4 Gas Turbine

3.1.5 H_2 and CH_4 buffers

The H_2 and CH_4 buffers are designed to temporarily store the produced hydrogen and methane, respectively. These two components are modelled in a very simple way, defining a mass balance equation.

- The H_2 buffer receives as input the mass flow rate of the produced hydrogen from the HTSE and sends as output the 90% of the produced hydrogen to the methanation reactor.
- The CH_4 buffer receives as input both the mass flow rate of the produced syngas from the methanation reactor and the mass flow rate required by the gas turbine to guarantee its setpoint load. The turbine's requested mass flow rate is an output of the turbine's model itself.

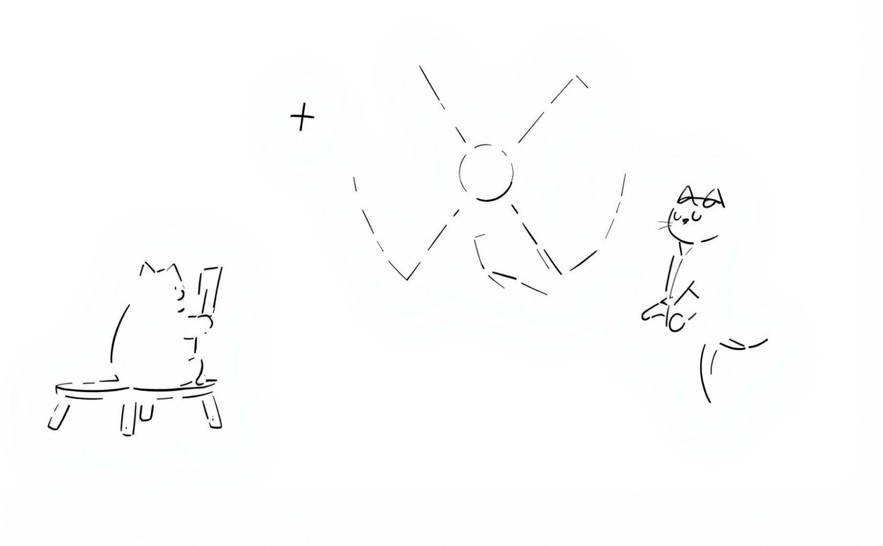


Figure 12: Simple schematic of the Methanation Reactor model in Dymola.

The instantaneous content of each buffer is then computed through a mass balance equation:

$$\frac{d}{dt}\text{Content} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} \tag{5}$$

As stated previously, the H_2 buffer is used to store the 10% of the produced hydrogen, which is then sold. The CH_4 buffer is used to store the syngas produced by the methanation reactor, which can later be used to cover the residual fossil fuel demand of the region.

3.1.6 Control Room

The Control Room is the central component of the plant model, responsible for managing the overall operation of the system. It defines the interconnections between the various components, such as the SMR, HTSE, Methanation Reactor, and Gas Turbine.

The Control Room takes as inputs:

- P_{nuc}: the hourly electrical power produced by the SMR.
- $P_{turbine}$: the hourly electrical power produced by the Gas Turbine.
- $Demand$: the hourly net grid demand, considering the contribution of the PV production.
- P_{HTSE}: the hourly electrical power consumed by the HTSE modules.

The Control Room is responsible for computing the Energy Balance of the system, defined as:

$$\text{EnergyExcess} = n_{\text{reactors}} \cdot P_{\text{nuc}} + P_{\text{turbine}} - P_{\text{HTSE}} \cdot n_{\text{HTSE}} - \text{Demand} \quad [\text{MW}] \tag{6}$$

The outputs of the Control Room are:

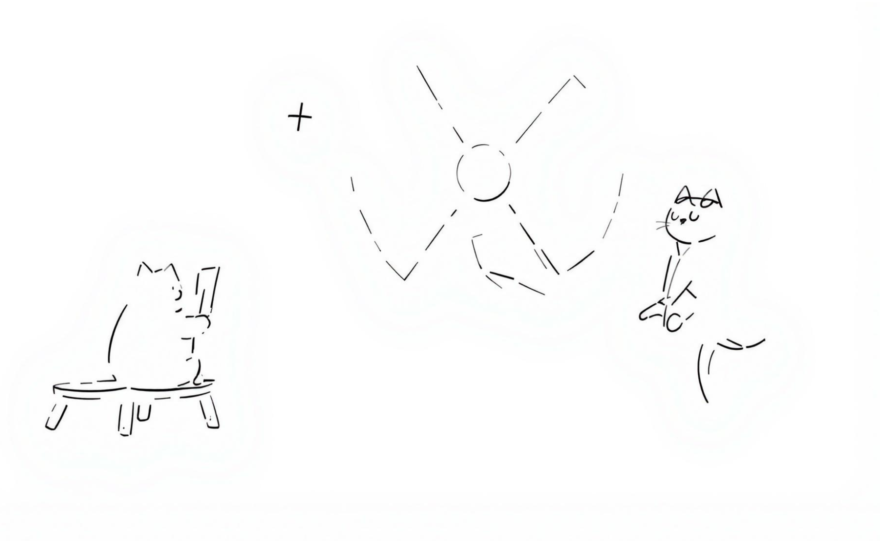


Figure 13: Simple schematic of the H_2 and CH_4 buffers in Dymola.

- $setpoint_{H_2}$: the setpoint for the hydrogen production rate.
- $turbine_load$: the setpoint for the turbine load (in percentage).
- $mflow_tapSteam$: the setpoint for the mass flow rate of steam extracted from the SMR's intermediate pressure tap.

Within the Control Room, the setpoints for the various components are imported as .csv files. Setpoints are generated beforehand by a Python script focused on the analysis of a single day demand profiles, as described in the next section. The Control Room also computes the instantaneous enegry excess based on the Energy Balance Equation, which will then be useful for the economic assessment.

4 METHODOLOGY

Using the Dassault Systèmes Dymola software [12], a production model was developed based on the object-oriented Modelica language.

Four representative scenarios were evaluated, as described in Section 1, each representing a 15-year operational phase of the plant, for a total time span of 60 years. The system is assumed to begin operation in 2025 and continue until 2085.

The optimal time horizon for dynamic simulation in this framework is set to 24 hours, in order to capture the daily variations in energy demand and the systems response to them.

The simulation methodology follows the steps outlined below:

1. For each scenario, an annual net electricity demand profile for the regional grid was derived through the data analysis developed in Section 2.

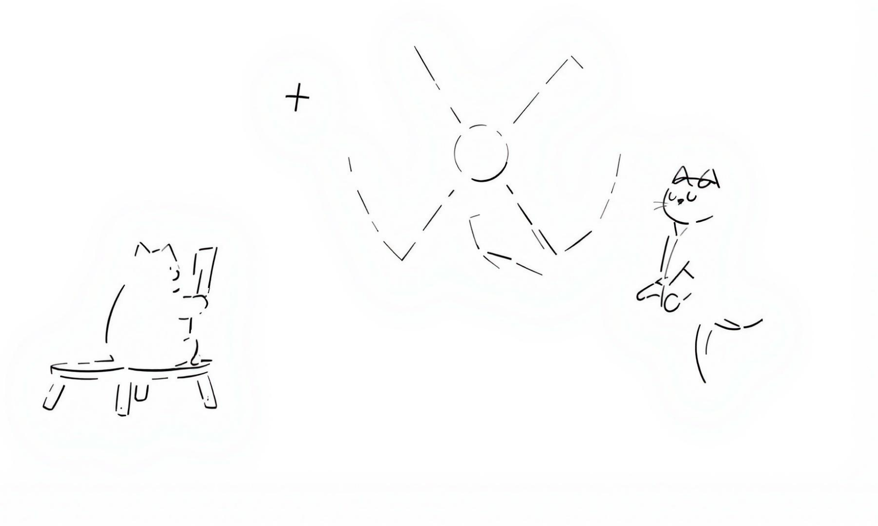


Figure 14: Control Room modelling in Dymola.

2. The annual profile was then clustered into five groups of days with similar load characteristics. These representative days were then dynamically simulated with Dymola.
3. A Python-based algorithm reconstructed the full annual profile by substituting actual days with the corresponding simulated cluster days.
4. Monthly energy balances — including electricity sold to the community, surplus energy exported to the market, energy deficits covered by purchases from the market — were then extracted and used as input for the economic analysis of the plant.
5. The yearly hydrogen and syngas production was also extracted from the simulation results, and added as input for the economic analysis.

4.1 Dymola simulation features

The Dymola model simulates the dynamic behaviour of the plant over a 24-hour period, but requires pre-processed input data to accurately account for all the relevant phenomena. The pre-processing methodology described below is based on the single-day demand profiles of the clusters previously obtained.

4.1.1 Model inputs

As highlighted in the Control Room subsection 3.1.6, the model requires, in addition to the hourly net grid demand (i.e., one of the cluster profiles), a set of daily setpoints for the following variables:

- Turbine load, constrained between 35% and 100% of the maximum load.
- HTSE load, constrained between 60% and 100% of the maximum load.

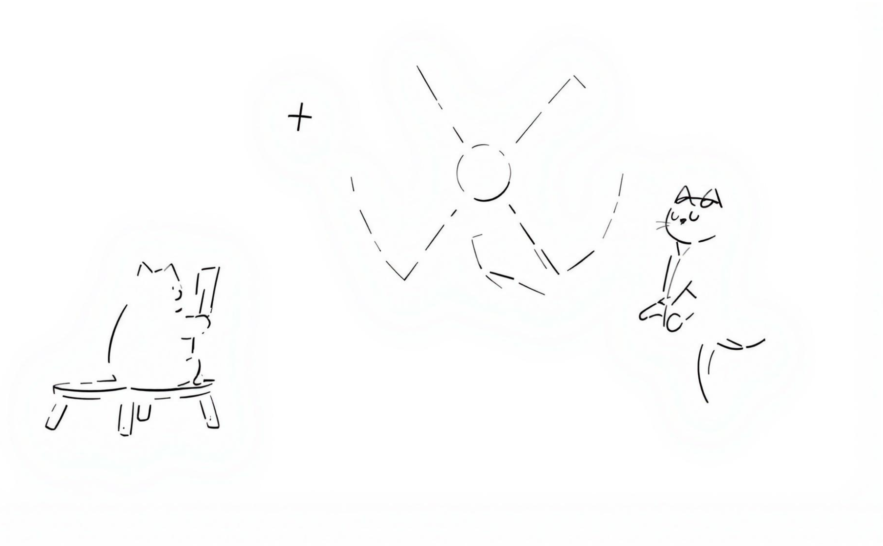


Figure 15: Example of annual net electricity demand profile for the regional grid.

- Steam tap flow from the intermediate pressure stage of the SMR.

Both the turbine and HTSE loads are subject to lower operational limits to avoid the risk of damaging the components when operated below their minimum thresholds.

4.1.2 Turbine and HTSE load setpoints

To evaluate the turbine and HTSE load setpoints, the proposed data pre-processing model first computes the electricity available for the hydrogen plant. This is done under the assumption that the SMRs operate continuously at their nominal power output, and that the gas turbine operates at its minimum load of 35%, ensuring it never runs below its allowed operational limit.

$$P_{e,\text{available}} = P_{e,\text{SMR}} + P_{\text{turbogas}}(35\%) - P_{e,\text{grid}} \tag{7}$$

This calculation helps determining the load profile of the H_2 plant, based on the load vs. electric power relationship described in the previous section, which was obtained by performing a sweep simulation of the HTSE model in Dymola.

$$\text{load}_{H_2} = \left(\frac{P_{e,\text{available}}}{214} + 0.197775\right) \cdot \frac{1}{N_{\text{HTSE}} \cdot 0.1} \tag{8}$$

Where the mass flow rate of hydrogen produced, $\dot{m}_{H_2} = \frac{P_{e,\text{available}}}{214} + 0.197775$, is divided by the nominal mass flow rate produced by one module (0.1 kg/s) multiplied by the number of modules N_{HTSE}.

The HTSE load profile is then clipped to remain within the operational range of 60% to 100%, where 60% represents the lower operational limit.

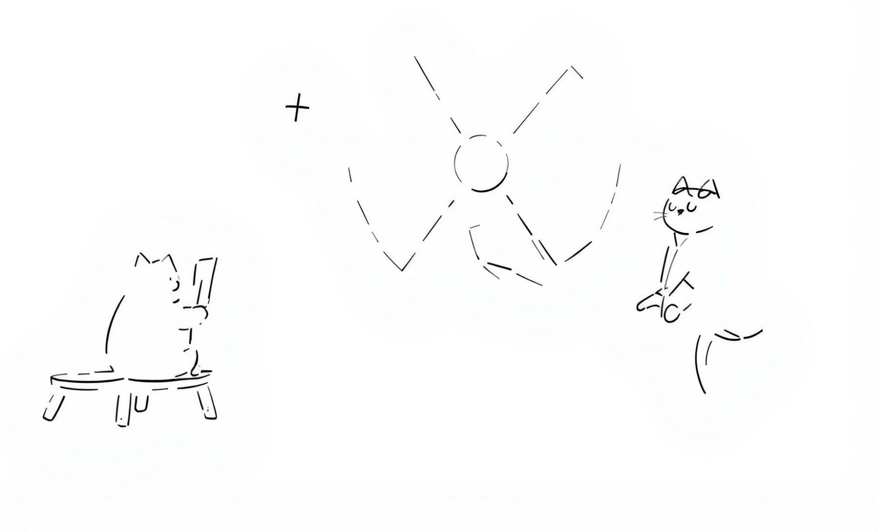


Figure 16: Example of clusters for an annual demand profile.

This clipped HTSE load profile subsequently defines the actual electricity consumption of the H_2 plant.

Based on the updated system energy balance, the turbine load profile is then determined as follows:

$$\text{load}_{\text{turbine}} + P_{e,\text{SMR}} = \text{HTSE}_{\text{consumption}} + P_{e,\text{grid}} \tag{9}$$

The turbine load must also be clipped. In this case, the lower limit was set to 60% instead of the physical minimum of 35%, in order to:

- Increase the overall electricity production available for sale.
- Avoid the more emission-intensive operation associated with lower load levels.

The resulting clipped turbine load profile, together with the actual consumption of the HTSE modules, determines the final electricity balance of the integrated plant.

The calculations show that, depending on the specific scenario, the system needs to import electricity from the grid in only 4% to 7% of the hours in a year, enabling highly efficient operation of the entire plant.

4.1.3 Rate limiting

Ramping up and ramping down rates for both the turbine and the HTSE were taken into account in the calculations.

According to the technical documentation of the selected gas turbine model, the ramping rates for the turbine are essentially unrestricted, as it can quickly reach its nominal power output.

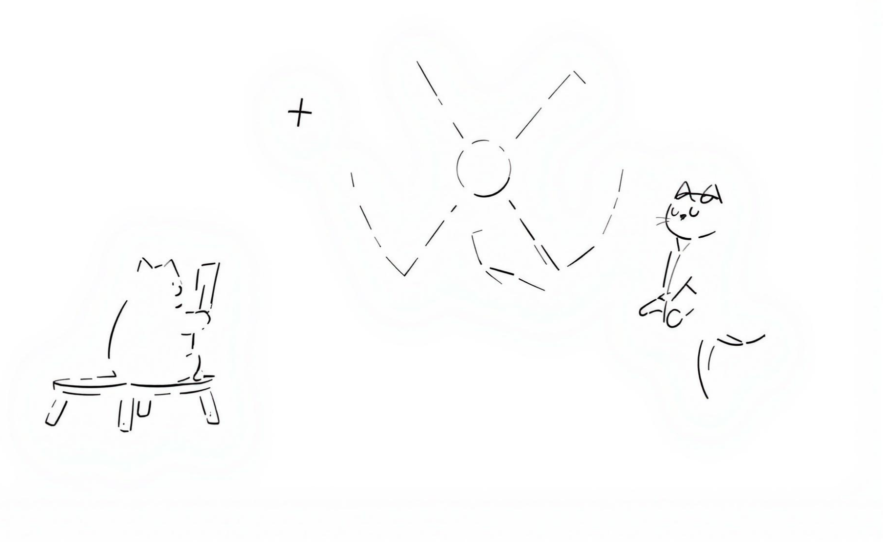


Figure 17: Example of final output results for 1-year operation.

In contrast, the HTSE module is subject to a ramping rate limit of 1% per hour. This constraint allows the model to reflect the physical limitations of the component, which cannot instantaneously adjust its load as the initial computed profile might suggest.

A Python function handles the ramping behaviour of both the turbine and HTSE loads, updating the previously calculated load profiles and corresponding energy balances accordingly.

4.1.4 Steam Tap Flow Setpoint

The steam tap flow setpoint is determined based on the HTSE load profile, computed as described above.

The relationship between the tap steam flowrate and the hydrogen production rate is established by performing a sweep simulation of the HTSE model in Dymola.

4.2 Results Post-Processing

The results of the Dymola simulation are post-processed to extract the monthly energy balances of the plant, which are then used for the economic analysis.

First, a `.csv` file containing the hourly simulation results is generated, and the full year is reconstructed as outlined in the previous section.

4.2.1 Energy Balances

Each annual electricity balance includes:

- Electricity sold to the community.

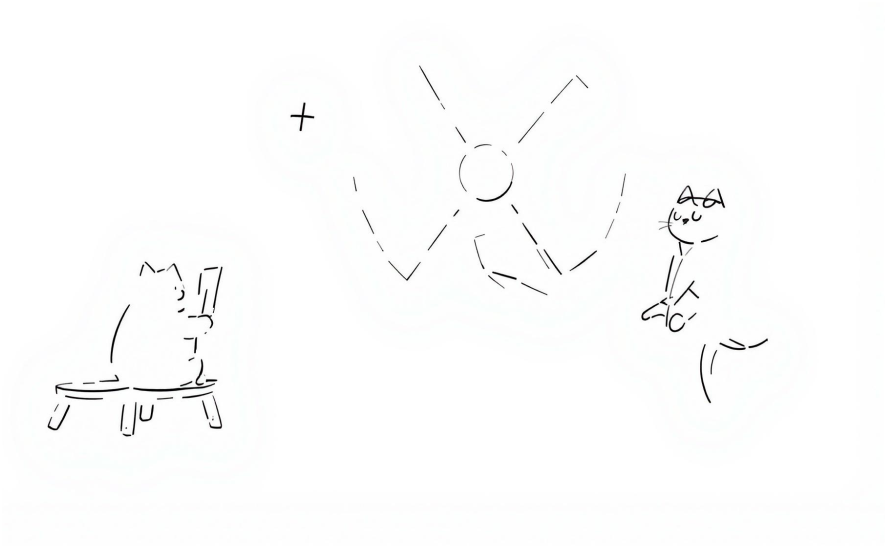


Figure 18: Example of 1-day load profiles for HTSE and gas turbine.

- Surplus energy exported to the market.
- Energy deficits covered by purchases from the market.
- Energy acquired from the community's photovoltaic excess production.

As the different scenarios progress over the plant's lifetime, a substantial increase is observed in the energy acquired from the community's photovoltaic excess production. This trend reflects the growing share of renewable energy sources in the region.

Monthly aggregate balances were then cumulatively computed to serve as input for the financial analysis.

4.2.2 Hydrogen and Syngas production

Hydrogen and syngas production data are also extracted from the simulation results and used to calculate the annual product balances.

Focusing on yearly balances simplifies the economic evaluation, as it assumes a stable selling price for these products throughout the year.

4.2.3 Hydrogen production

The yearly hydrogen production is calculated by summing the daily production values over the entire year.

The total hydrogen production is expressed in energy terms (MWh), using the lower heating value (LHV) of hydrogen, which is approximately 36.11 kWh/kg.

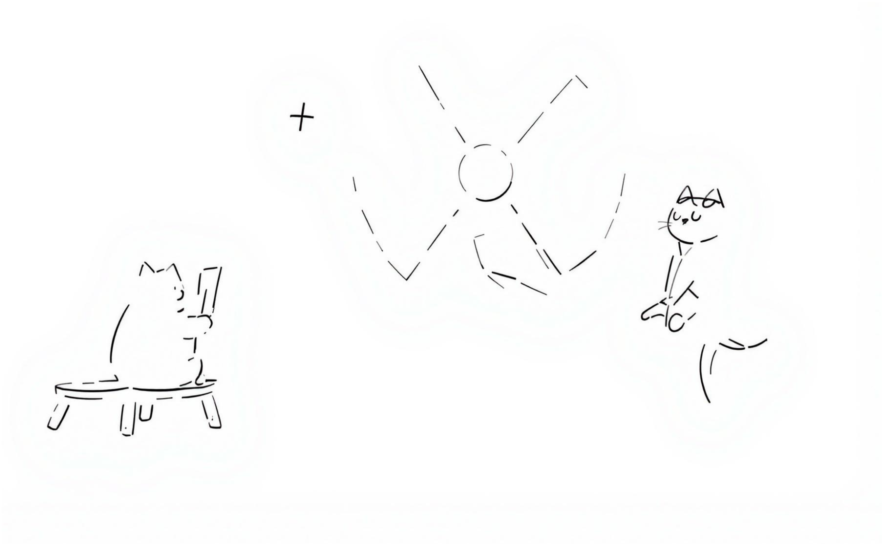


Figure 19: Example of 1-year energy balance values.

4.2.4 Syngas production

The stored syngas is assumed to contribute towards meeting the community remaining fossil fuel demand, thereby supporting the region decarbonization efforts.

Accordingly, the yearly syngas production is converted into energy units (MWh) using the lower heating value (LHV) of the produced gas, which is approximately 57.4539 MJ/kg.

5 ECONOMIC STUDY

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